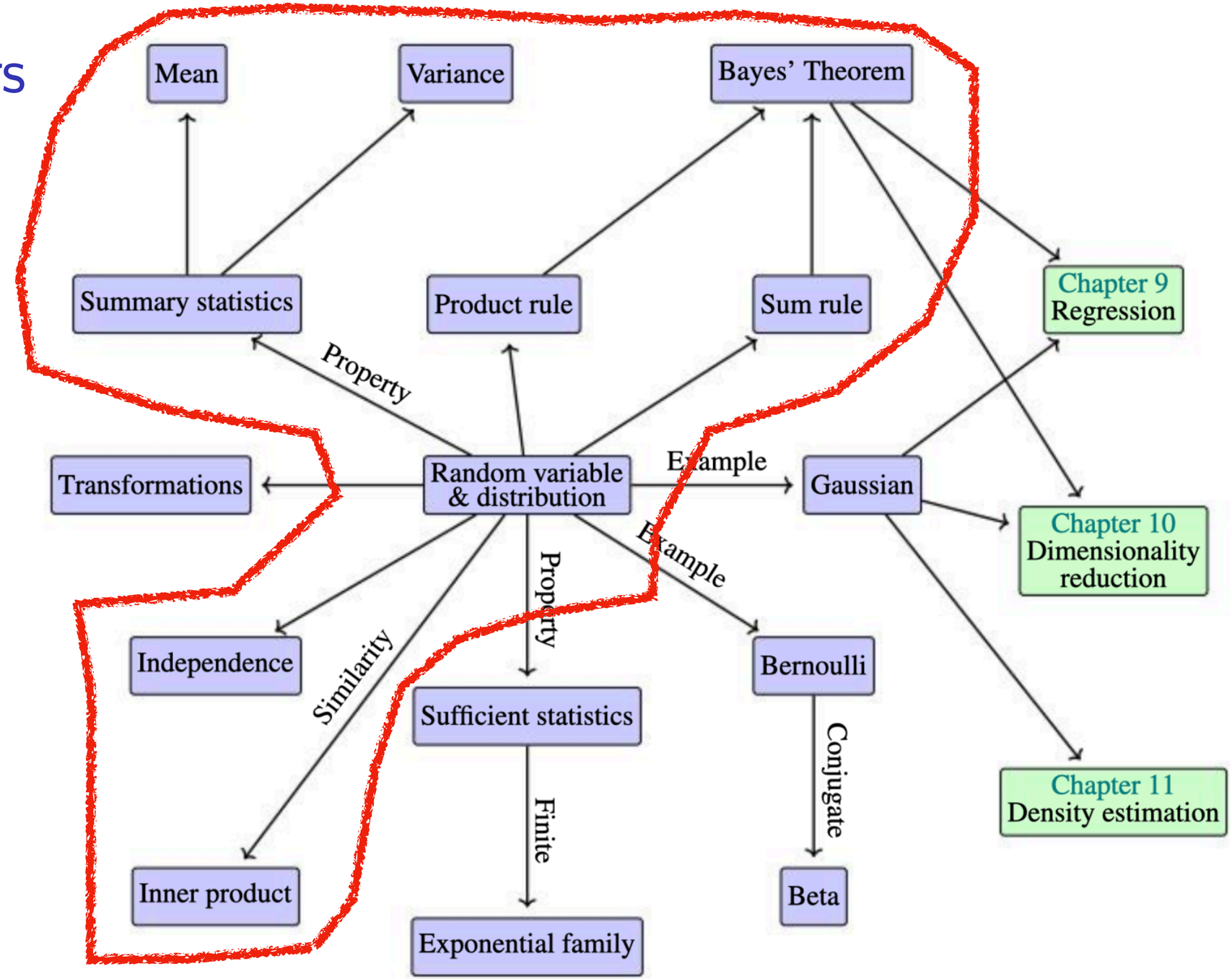
Seminar 2 - Probabilities

ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

January 14, 2026

This Seminar Covers



Why probability space?

Why random variables?

Why distributions?

Sample and Event Spaces

Sample Space Ω:

Set of all possible outcomes of the experiment; for example, space of two successive coin tosses: $\{hh, tt, ht, th\}$

Event Space $\mathscr{A}$:

Space of potential results of the experiment; subset A *of the sample space* Ω *is in the event space* $\mathscr{A}$*;*

in the experiment we can observe if $w \in \Omega$ *is in* A*.*

Target Space

Target Space $\mathscr{T}$:

Its elements are states; there is a function

$$\boldsymbol{X} : \boldsymbol{\Omega} \rightarrow \mathscr{T}$$

that takes an element of $\boldsymbol{\Omega}$ and returns a particular quantity of interest x, a value in $\mathscr{T}$;

This mapping $\boldsymbol{X}$ is called random variable.

Random Variable

Random Variable $X : \Omega \to \mathscr{T}$

A random variable X is a mapping from a set of outcomes Ω of an experiment to a set $\mathscr{T} \subset \mathbb{R}$ of values.

When $\mathscr{T}$ is a finite or countably infinite set, we call X a *discrete* random variable.

When $\mathscr{T}$ is an uncountable subset of $\mathbb{R}$, we call X a *continuous* random variable.

Random Variable Examples

Ex1: Flip of a coin.

Sample space $\Omega = \{h, t\}$ *Target space* $\mathscr{T} = \{0,1\}$

Random variable $\boldsymbol{X} = \begin{cases} 0, & t \\ 1, & h \end{cases}$

Ex2: Tossing two coins and *counting the number* of heads.

Sample space $\Omega = \{hh, tt, ht, th\}$ *Target space* $\mathscr{T} = \{0,1,2\}$

Random variable $\boldsymbol{X} = \begin{cases} 0, & tt \\ 1, & th \text{ or } ht \\ 2, & hh \end{cases}$

What is the *probabilities* of these scenarios happening?

Probability Space

Probability Space P:

For each event $A \in \mathscr{A}$, we have a number $P(A)$ that measures the probability or degree of belief that the event will occur.

For a random variable $\boldsymbol{X}$,

$$P(\boldsymbol{X} = x) \in [0,1]$$

Probability Mass Functions

A *discrete* random variable is fully characterized by its *probability mass function (pmf)*.

$$\mathscr{T} \text{ is discrete.}$$

For $x \in \mathscr{T}$*, the pdf is*

$$P(\boldsymbol{X} = x) \geq 0$$

and

$$\sum_{x \in \mathscr{T}} P(\boldsymbol{X} = x) = 1$$

Probability Density Functions

A *continuous* random variable is fully characterized by its *probability density function (pdf)*.

$$\mathscr{T} \text{ is continuous, on a real line } \mathbb{R}.$$

For $x \in \mathscr{T}$*, the pdf is*

$$p(\boldsymbol{X}) \geq 0$$

and

$$\int_{x \in \mathbb{R}} p(x)dx = 1$$

Cumulative distribution function is $p(\boldsymbol{X} \leq x) = \int_{-\infty}^{x} p(x)dx.$

Cumulative Distribution Function

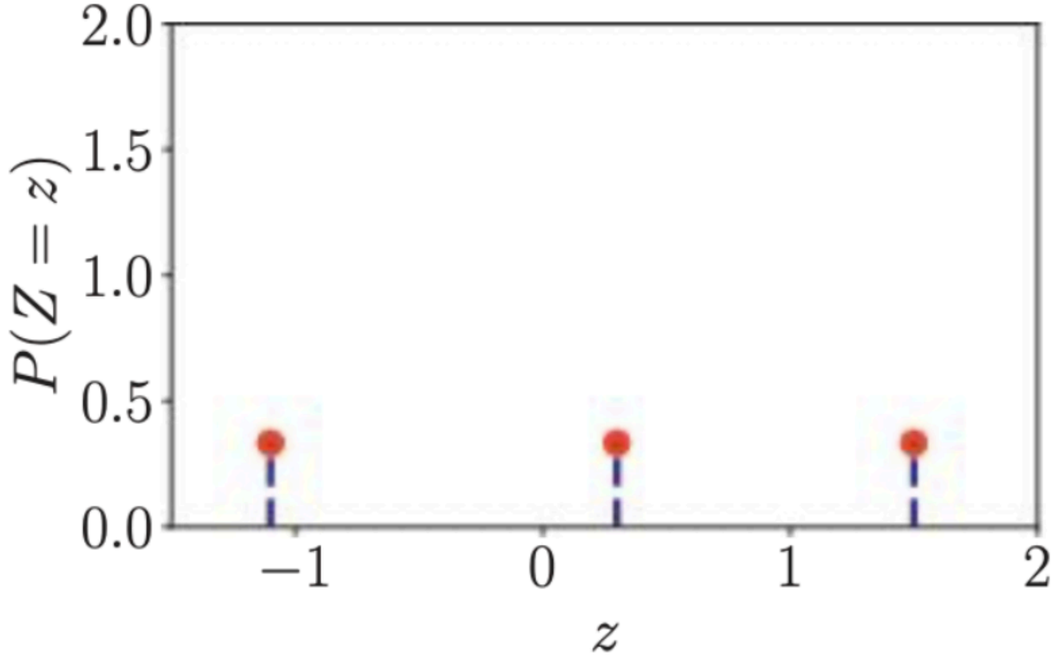
A *cumulative distribution function* (cdf) of a multivariable real-valued random variable X with states $\boldsymbol{x} \in \mathbb{R}^D$ is given by

$$F_X(\boldsymbol{x}) = P(X_1 \leq x_1, \ldots, X_D \leq x_D)$$

where $X = [X_1, \ldots, X_D]^\top$ and $\boldsymbol{x} = [x_1, \ldots, x_D]^\top$.

Example: Uniform Distribution for Discrete and Continuous RVs.

Discrete



Continuous

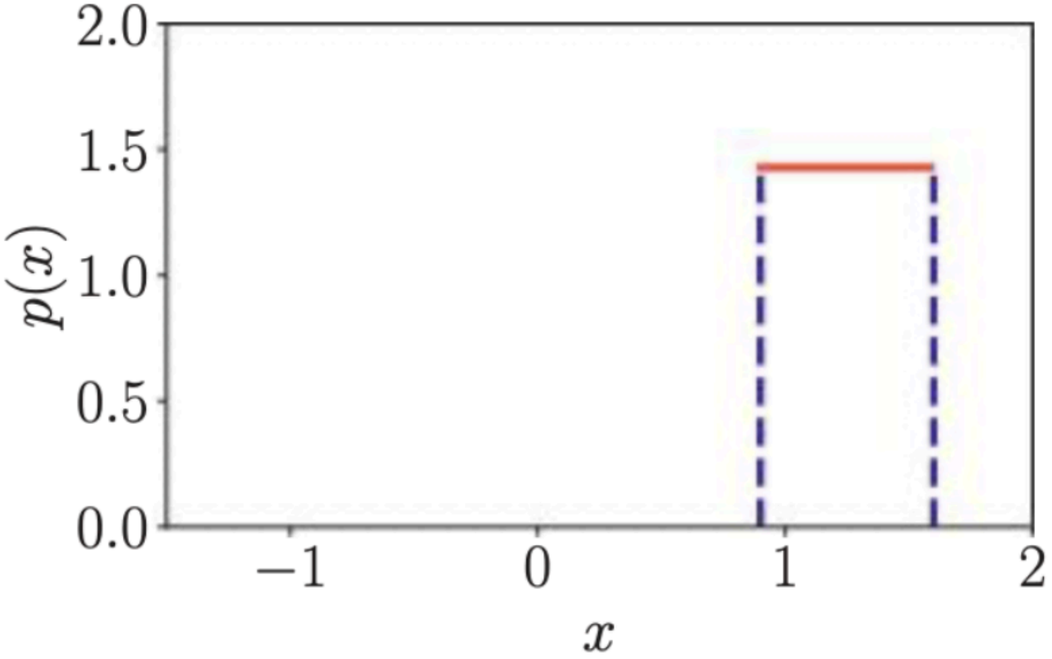


figure from Mathematics for Machine Learning textbook.

Probabilities

Joint probability $p(x, y)$

Marginal probability $p(x)$

Conditional probability $p(x \mid y)$

Note:

univariate distribution when random variable is scalar.

multivariate distribution when we have a vector of random variables.

Joint Probability

A vector random variable is fully characterized by the joint pdf/pmf of its components

Discrete case:

$$P(\boldsymbol{X}_1 = x_1, \boldsymbol{X}_2 = x_2, \ldots, \boldsymbol{X}_n = x_n) \geq 0$$

$$\sum_{x_1} \cdots \sum_{x_n} P(\boldsymbol{X}_1 = x_1, \boldsymbol{X}_2 = x_2, \ldots, \boldsymbol{X}_n = x_n) = 1$$

Continuous case:

$$P_{\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n}(x_1, x_2, \ldots, x_n) \geq 0$$

$$\int_{x_1} \cdots \int_{x_n} P_{\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n}(x_1, x_2, \ldots, x_n) dx_1 dx_2 \cdots dx_n = 1$$

Marginalization (Sum Rule)

Example: $$p(\boldsymbol{X} = x, \boldsymbol{Y} = y) = \frac{n_{ij}}{N}$$

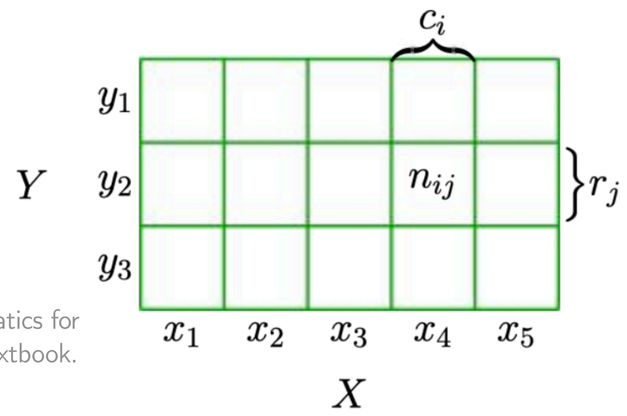


figure from Mathematics for Machine Learning textbook.

$$p(\boldsymbol{X} = x) = \frac{c_i}{N} = \frac{\sum_{j=1}^{J} n_{ij}}{N}$$

$$p(\boldsymbol{Y} = y) = \frac{r_j}{N} = \frac{\sum_{i=1}^{I} n_{ij}}{N}$$

To get the marginalized probability,

$$p(x) = \begin{cases} \sum_{y \in \mathcal{Y}} p(x, y) & \text{if } y \text{ is discrete} \\ \int_{\mathcal{Y}} p(x, y) dy & \text{if } y \text{ is continuous} \end{cases}$$

How Can Joint, Marginal, and Conditional Probabilities Relate?

Product rule:

$$p(x, y) = p(y \mid x)p(x)$$

Bayes' rule:

posterior *likelihood* *prior*

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}$$

evidence

https://youtu.be/PLAZyu73tWE?si=yVVa1iFToQRUuUcF&t=620

$$p(y) := \int p(y \mid x)p(x)dx = \mathbb{E}_X[p(y \mid x)]$$

Expectation

Let g be a function from $\mathscr{T}$ to $\mathbb{R}$, we define the expectation $\mathbb{E}$ as

Discrete case:

$$\mathbb{E}_{\boldsymbol{X}}[g(x)] = \sum_{x \in \mathscr{X}} g(x) P(\boldsymbol{X} = x)$$

Continuous case:

$$\mathbb{E}_{\boldsymbol{X}}[g(x)] = \int_{\mathscr{X}} g(x) p(x) dx$$

multivariate (D-dim):

$$\mathbb{E}_{\boldsymbol{X}}[g(\boldsymbol{x})] = \begin{bmatrix} \mathbb{E}_{\boldsymbol{X}}[g(x_1)] \\ \vdots \\ \mathbb{E}_{\boldsymbol{X}}[g(x_D)] \end{bmatrix}$$

where $\mathscr{X}$ is the set of possible outcomes (the target space).

Expected Value is a Linear Operator

Linear operator $\mathscr{A}$:

$$\mathscr{A}(g(\boldsymbol{x})) = m_g \qquad \mathscr{A}(a \cdot g(\boldsymbol{x})) = a \cdot m_g$$

$$\mathscr{A}(h(\boldsymbol{x})) = m_h \quad \longrightarrow \quad \mathscr{A}(g(\boldsymbol{x}) + h(\boldsymbol{x})) = m_g + m_h$$

Given a real-valued function $f(\boldsymbol{x}) = ag(\boldsymbol{x}) + bh(\boldsymbol{x})$ *where* $a, b \in \mathbb{R}$, *and* $\boldsymbol{x} \in \mathbb{R}^D$, *we obtain*

$$\mathbb{E}_{\boldsymbol{X}}[f(\boldsymbol{x})] = a\mathbb{E}_{\boldsymbol{X}}[g(\boldsymbol{x})] + b\mathbb{E}_{\boldsymbol{X}}[h(\boldsymbol{x})]$$

Mean

Special case of *Expectation* by choosing g to be *identity* function.

Mean of a random variable $\boldsymbol{X}$ with states $\boldsymbol{x} \in \mathbb{R}^D$ is given by

$$\mathbb{E}_{\boldsymbol{X}}[\boldsymbol{x}] = \begin{bmatrix} \mathbb{E}_{\boldsymbol{X}}[x_1] \\ \vdots \\ \mathbb{E}_{\boldsymbol{X}}[x_D] \end{bmatrix}$$

where

$$\mathbb{E}_{x_d}[x_d] := \begin{cases} \int_{\mathcal{X}} x_d \, p(x_d) \, dx_d, \textit{ if } \boldsymbol{X} \textit{ is continous} \\ \sum_{x_i \in \mathcal{X}} x_i \, p(x_d = x_i), \textit{ if } \boldsymbol{X} \textit{ is discrete} \end{cases}$$

Taking weighted average of the probability distribution

Mean vs. Median vs. Mode

Median

Middle value if we sort the values, i.e., 50% of the values are greater than the median and 50% are smaller than the median.

Mode

Most frequency occurring value.

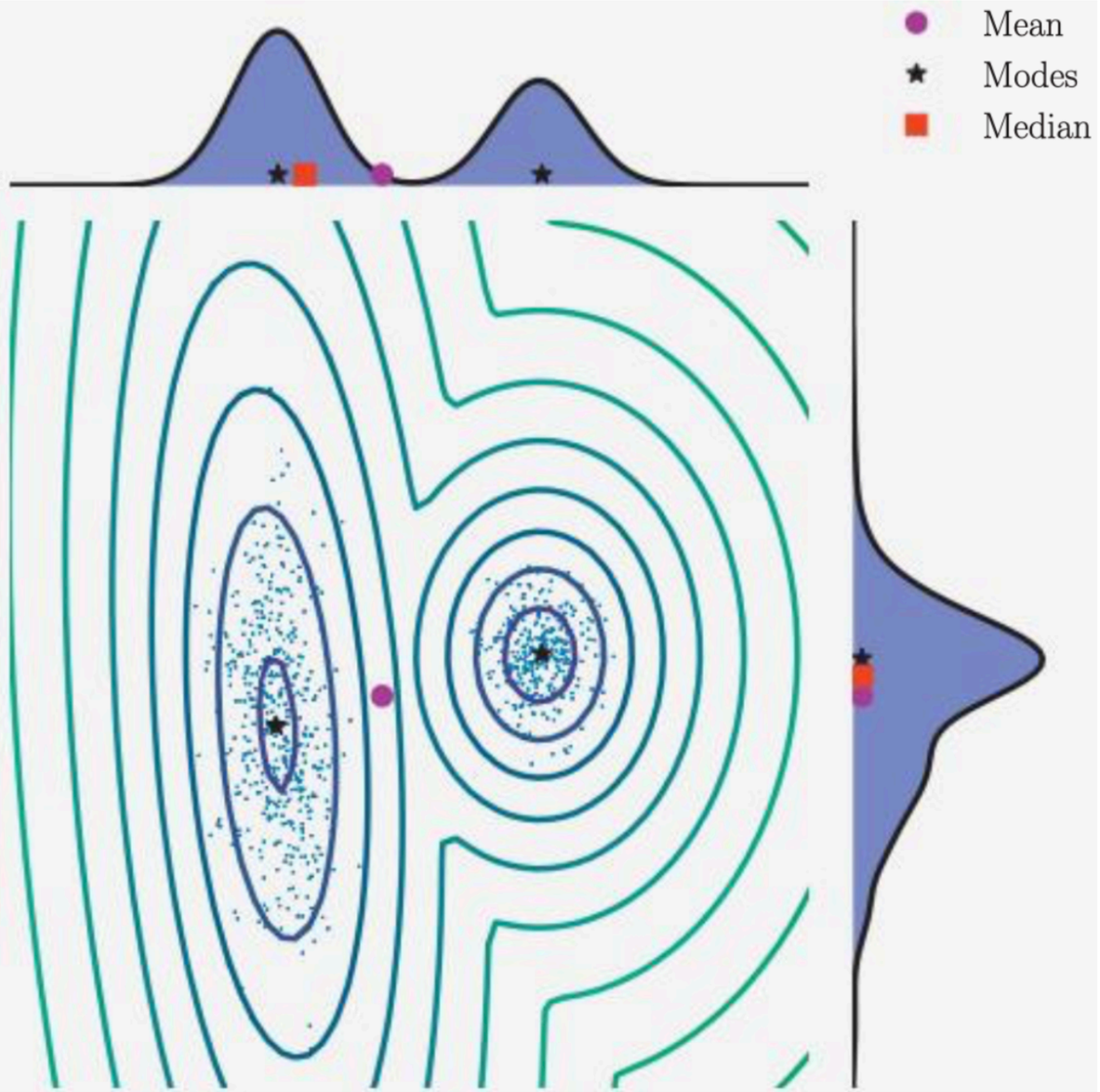


figure from Mathematics for Machine Learning textbook.

How Dependent Random Variables Are to One Another?

Covariance

Between two univariate random variables $X, Y \in \mathbb{R}$ is given by the expected product of their deviations from their respective means.

$$\mathrm{Cov}_{X,Y}\left[x, y\right] := \mathbb{E}_{X,Y}\left[(x - \mathbb{E}_X[x])(y - \mathbb{E}_Y[y])\right]$$

Remark.

$$\mathrm{Cov}_{X,Y}\left[x, y\right] := \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

Covariance of a Variable with Itself

Variance

$$\mathbb{V}_X\,[x] := \mathrm{Cov}_X\,[x, x]$$

$$\mathbb{V}_X\,[x] = \mathbb{E}_X[(x-\mu)^2]$$

where $\mu = \mathbb{E}_X[x]$.

$$\mathbb{V}_X\,[x] = \mathbb{E}_X[x^2] - (\mathbb{E}_X[x])^2$$

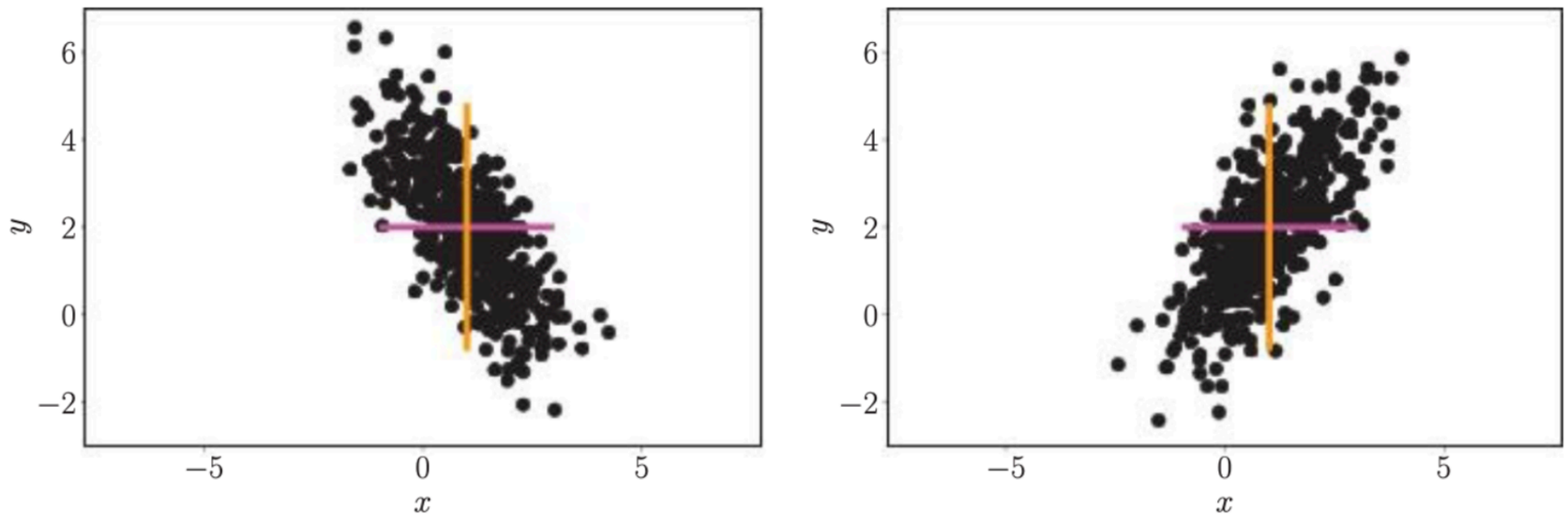
raw-score formula for variance

Finding Relation between Two Random Variables

Correlation

between two random variables $X, Y \in \mathbb{R}$ is given by

$$\mathrm{Corr}[x, y] := \frac{\mathrm{Cov}[x, y]}{\sqrt{\mathbb{V}[x]\mathbb{V}[y]}} \in [-1,1]$$



Identical mean and variance.

Different correlation

figure from Mathematics for Machine Learning textbook.

Empirically Compute Mean and Covariance

Empirical Mean

$$\bar{\boldsymbol{x}} := \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \qquad D\text{-dimensional}$$

Empirical Covariance

$$\Sigma := \frac{1}{N}\sum_{n=1}^{N} (\boldsymbol{x}_n - \bar{\boldsymbol{x}})(\boldsymbol{x}_n - \bar{\boldsymbol{x}})^\top \qquad D \times D$$

where $\boldsymbol{x} \in \mathbb{R}^D$.

Intuition of Variance

Sum of pairwise differences between all pairs of observations.

$$\frac{1}{N^2}\sum_{i,j=1}^{N}(x_i - x_j)^2$$

$$2[\frac{1}{N}\sum_{i=1}^{N}x_i^2 - (\frac{1}{N}\sum_{i=1}^{N}x_i)^2]$$

This is twice the raw-score expression.

Express the sum of pairwise distance (N^2 of them),

as a sum of deviations from the mean (N of them).

Equivalence between the pairwise distances and the distances from the centre of the set of points.

Sums and Transformations of Random Variables

Consider two random variables X, Y with states $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^D$. Then,

$$\mathbb{E}[\boldsymbol{x} + \boldsymbol{y}] = \mathbb{E}[\boldsymbol{x}] + \mathbb{E}[\boldsymbol{y}]$$

$$\mathbb{E}[\boldsymbol{x} - \boldsymbol{y}] = \mathbb{E}[\boldsymbol{x}] - \mathbb{E}[\boldsymbol{y}]$$

$$\mathbb{V}[\boldsymbol{x} + \boldsymbol{y}] = \mathbb{V}[\boldsymbol{x}] + \mathbb{V}[\boldsymbol{y}] + \mathrm{Cov}[\boldsymbol{x}, \boldsymbol{y}] + \mathrm{Cov}[\boldsymbol{y}, \boldsymbol{x}]$$

$$\mathbb{V}[\boldsymbol{x} - \boldsymbol{y}] = \mathbb{V}[\boldsymbol{x}] + \mathbb{V}[\boldsymbol{y}] - \mathrm{Cov}[\boldsymbol{x}, \boldsymbol{y}] - \mathrm{Cov}[\boldsymbol{y}, \boldsymbol{x}]$$

Statistical Independence

Two random variables X, Y are *statistically independent* if and only if

$$p(\boldsymbol{x}, \boldsymbol{y}) = p(\boldsymbol{x})p(\boldsymbol{y})$$

Intuition: the value of $\boldsymbol{y}$ (once known) does not add any additional information about $\boldsymbol{x}$ (and vice versa). Hence,

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = p(\boldsymbol{y}) \qquad \mathbb{V}_{X,Y}[\boldsymbol{x} + \boldsymbol{y}] = \mathbb{V}_X[\boldsymbol{x}] + \mathbb{V}_Y[\boldsymbol{y}]$$

$$p(\boldsymbol{x} \mid \boldsymbol{y}) = p(\boldsymbol{x}) \qquad \mathrm{Cov}_{X,Y}[\boldsymbol{x}, \boldsymbol{y}] = \boldsymbol{0}$$

Conditional Independence

Two random variables X and Y are *conditionally independent* given Z if and only if

$$p(\boldsymbol{x}, \boldsymbol{y} \mid \boldsymbol{z}) = p(\boldsymbol{x} \mid \boldsymbol{z}) p(\boldsymbol{y} \mid \boldsymbol{z}) \quad \textit{for all} \quad \boldsymbol{z} \in \mathcal{Z}$$

$$X \perp\!\!\!\perp Y \mid Z$$

Intuition: given that we know $\boldsymbol{z}$, the knowledge about $\boldsymbol{y}$ does not our knowledge of $\boldsymbol{x}$.

Geometric Properties of Random Variables

Think of random variables as vectors in a vector space.

(I) If we define

$$\langle X, Y \rangle := \mathrm{Cov}[x, y]$$

for *zero mean* random variables X and Y, we obtain an inner product.

(II) The length of a random variable is

$$\|X\| = \sqrt{\mathrm{Cov}[x, x]} = \sqrt{\mathscr{V}[x]} = \sigma[x]$$

A random variable with length 0 is deterministic.

(III) The angle between two random variables is the correlation.

$$\cos\theta = \frac{\langle X, Y \rangle}{\|X\| \, \|Y\|} = \frac{\mathrm{Cov}[x, y]}{\sqrt{\mathscr{V}[x]\mathscr{V}[y]}}$$

What is the Inner *Product* between Two *Statistically Independent* Random Variables?

$$X \perp\!\!\!\perp Y \Leftrightarrow \langle X, Y \rangle = 0$$

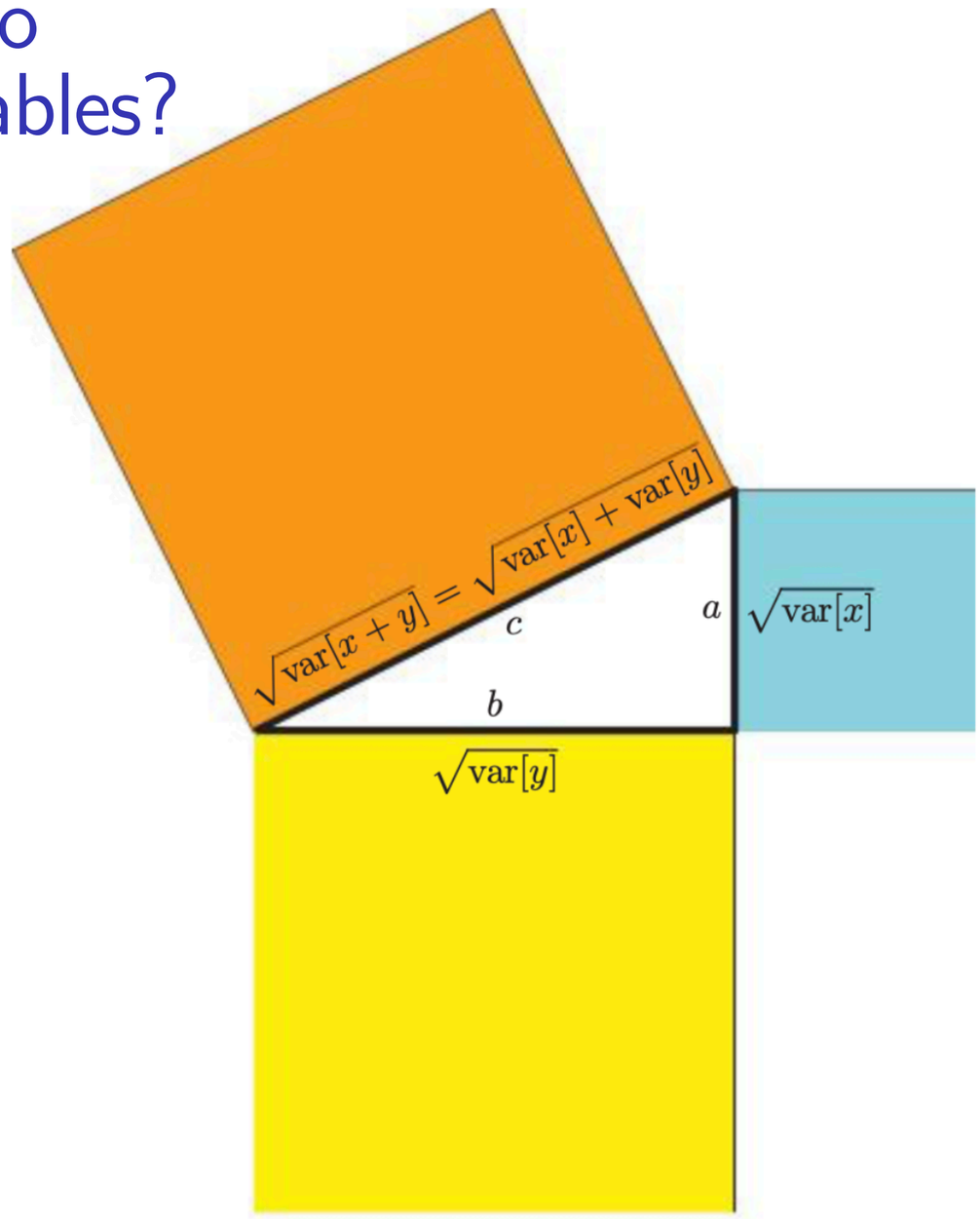


figure from Mathematics for Machine Learning textbook.

Takeaways

AI foundations even to this date is based on *probability*.

A blog on generative diffusion models (AI).

https://lilianweng.github.io/posts/2021-07-11-diffusion-models/